

Lab 1: Knowledge is Power (KIP)

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Table of Contents

1 Introduction 2

2 KIP (Knowledge Is Power) Product Description 3

 2.1 Key Product Features and Capabilities3

 2.2 Major Components (Hardware/Software).....5

 2.2.1 Presentation Layer..... 5

 2.2.2 Application Layer 6

 2.2.3 Database Layer..... 8

 2.2.4 Hardware Layer..... 8

3 Prototype Placeholder 9

4 Target Audience..... 9

5 Glossary 10

6 References 11

List of Figures

Figure 1: 2017 Legal Literacy Complication Statistics 2

Figure 2: Major Functional Components Diagram..... 5

Figure 3: Placeholder Image 9

List of tables

No table of tables found.

1. Introduction

Knowledge Is Power, or KIP for short, is a browser-based legal literacy learning game platform designed to encourage young adults to learn about the legal system. KIP will be equipped with game modules that test the users' learning through quizzes, tests, and immediate feedback. KIP does not act as a legal certification system, but just as a mode of education reinforcement. It seeks to supplement the legalese taught within the standardized education system.

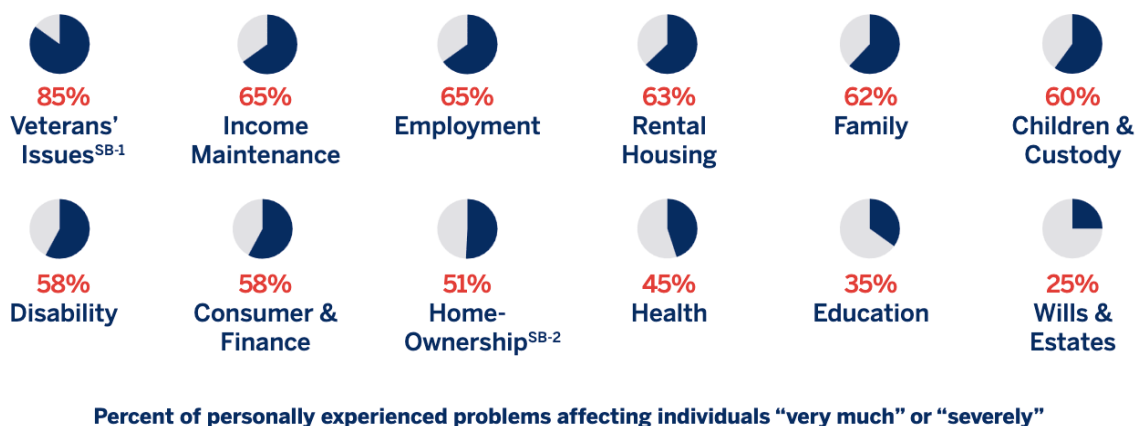


Figure 1: 2017 Legal Literacy Complication Statistics

Young people ages 16–19 increasingly encounter laws involving driving, digital conduct, work, school, ownership, contracts, and safety without fully understanding their inner workings and the consequences that may ensue. As they gain more independence into adulthood, this lack of legal awareness can lead to poor decisions and serious consequences. Despite this, law-related education remains largely absent from K-12 curricula, leaving young adults underprepared for

real-world legal situations. Additionally, teen drivers face higher safety risks that are often linked to poor judgement and decision-making. [6.1, 6.2, 6.3, 6.4]

The platform should teach legal concepts through interactive activities that ask users to apply laws to realistic situations rather than simply memorize definitions. To remain accessible, it should use clear and consistent language, concise explanations, mobile-friendly web design, and low-stakes practice games that students can complete independently. It should also explain the connection between each lesson and its official legal sources while introducing legal vocabulary in context so that users can recognize practical terms used across different legal topics.

2. KIP (Knowledge Is Power) Product Description

KIP is structured as a web-based educational service built around learning modules, interactive games, user progress data, and source-backed legal content. The system presents users with short misconception-based quizzes and longer branching scenarios, then uses their performance to recommend future practice. Its software components support account access, scenario delivery, feedback generation, content curation, source referencing, and persistent progress tracking. The goal of these features is to assess users' current legal understanding, strengthen their recognition of legal situations, and reinforce informed decision-making through repeated low-stakes practice.

2.1. Key Product Features and Capabilities

KIP uses focused games, real-world examples, curated media, and official resources to educate young adults on practical legal boundaries. It aims to assess, improve, and reinforce legal knowledge among tech-friendly youth through repeatable, low-stakes interactive activities.

Its user experience (UX) is designed to increase recognition of legal situations, improve decision-making skills, and strengthen long-term retention of legal concepts. One game KIP will utilize is “Fact or Fiction?”.

This short-form game presents users with common legal beliefs and misconceptions, offering “true”, “false”, or “context-dependent” as answers. This approach will help users challenge assumptions, spot legal nuance, and build foundational knowledge to prepare them for longer and more elaborate scenarios. Another game KIP will employ is “Branching Narrative”.

This multiple-choice, long-form game encourages users to interact with people, objects, and documents to learn scenario details presented to them before deciding. Every choice will demonstrate its consequences to illuminate its legal significance, which encourages users to slow down, evaluate context, and understand what information an outcome is dependent on. KIP tracks topical strengths, weaknesses, and completion during use. This information allows KIP to recommend relevant games and modules while helping each game session offer questions, scenarios, and legal concepts that meaningfully test the user.

2.2. Major Components (Hardware/Software)

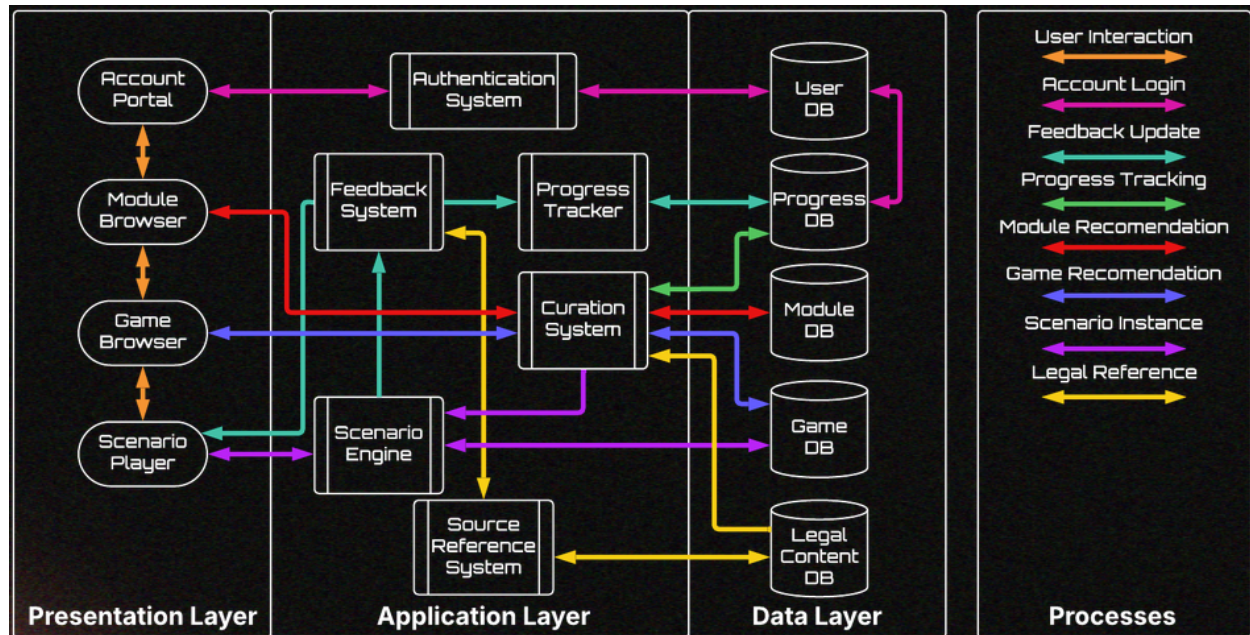


Figure 2: Major Functional Components Diagram

Knowledge Is Power will be comprised of four main layers: presentation, application, data, and hardware.

2.2.1 Presentation Layer

For the presentation layer, the tech stack will include JavaScript, Bootstrap, and HTML5. These technologies, in combination, will allow KIP to provide a user-friendly and efficient navigable platform for interaction.

2.2.1.1 Account Portal System

The account portal system allows established users to log in, and newcomers to create an account or play games without an account to get a taste of KIP, albeit with limited features.

2.2.1.2 Module Browser System

The module browser displays available legal learning modules such as traffic law, online conduct, school and work, and personal safety. It allows users to select a topic and view their progress status for that topic's module.

2.2.1.3 Game Browser System

The game browser displays the available games and prioritizes recommendations based on the user's progress, performance history, and demonstrated strengths or weaknesses in the legal subtopics covered by each game.

2.2.1.4 Scenario Player

To conclude KIP's application layer, the scenario player is the sole interface users interact with while playing KIP's games. It presents interactive questions, branching scenarios, feedback, and source references during gameplay.

2.2.2 Application Layer

The application layer tech stack includes Java and JDBC. KIP will engage an authentication system, scenario engine, feedback system, curation system, and source reference system.

2.2.2.1 Authentication System

The authentication system manages user login, account creation, guest access, and session validation. It determines whether a user is using KIP as a registered account or as a limited guest, and it controls which features are available based on that status. The scenario

engine loads and manages the interactive content used during KIP's games. For a play session, it assembles a "match" of play, with multiple "rounds." Each round establishes unique choices, scenario branches, consequences, and game states that appear in sequence. This is primarily informed by two fed selected game and the user's previous actions.

2.2.2.2 Feedback System

The feedback system provides explanations after user decisions. It identifies whether an answer is correct, incorrect, or context-dependent, explaining the legal reasoning behind the result to help the user understand why a specific choice was safe, risky, or incomplete.

2.2.2.3 Curation System

The curation system organizes and recommends learning content based on the user's progress and performance history. It determines what games, modules, and scenarios to prioritize, depending on where the user needs more practice, while also allowing users to freely browse and ultimately select any available content.

2.2.2.4 Source Reference System

The source reference system connects game content and feedback to official legal sources, educational resources, and relevant vocabulary. It allows users to see where KIP's information comes from and, when desired, review the original legal language and documents behind a lesson. It also supports content maintenance by tracking changes to legislation, updating references when needed, and flagging KIP content that may be outdated so administrators can review it, revise it, or disable access to it.

2.2.3 Database Layer

The database data layer will primarily consist of SQLite. The databases that will be developed consist of user, progress, module game, and legal content. The functions of the databases are to store user account information; stores user progress records; stores learning modules and associated content; stores game scenario questions and gameplay information; stores legal and educational resources.

2.2.4 Hardware Layer

The hardware layer will be split between server-side and client-side components. The former, server-side will consist of a virtual private that hosts the Java application, manages server-side processing, and provides access to the SQLite database file and application resources. The latter, client-side, will be comprised of consumer devices capable of accessing a browser-based web address through an internet connection. The functions of the hardware layer include running Java application code, storing persistent data, processing user requests, and delivering the web-based interface to users.

3. Prototype Placeholder



Figure 3: Placeholder Image

4. Target Audience

KIP's primary target audience is young people aged 16–19 who are preparing to enter adulthood and navigate high-risk legal areas, such as driving, work, and substance use. This group is well suited for KIP because its members are old enough for legal consequences to become personally significant, but still young enough to be especially receptive to education, habit formation, and the correction of misconceptions. By focusing on users at this transitional stage, KIP can help build foundational legal awareness and encourage informed decision-making before risky habits or misunderstandings become more difficult to change. The secondary stakeholders include parents, teachers, and high school administrators that use KIP as a supplemental educational tool. These stakeholders benefit from KIP to reinforce legal awareness outside the classroom. Future users of KIP may be members of juvenile diversion programs, community youth groups, and driving schools. These groups may use KIP as educational resources to improve legal literacy in structured settings.

5. Glossary

CDC: Centers for Disease Control and Prevention

KIP: Knowledge Is Power

MFCD: Major Functional Component Diagram

UX: User Experience

6. References

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